

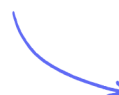
Non Calculator Questions

Exponentials & Logarithms

Exponential Growth & Decay / Logarithms

Medium (6 questions)	/8
Hard (4 questions)	/10
Very Hard (3 questions)	/13
Total Marks	/31

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Medium Questions

1 $\log_a 64 = 2$

Write down the value of a .

(1 mark)

2 Find the value of $\log_2 16$

(1 mark)

3 $y = 10^x$

Write x in terms of y .

$x = \dots\dots\dots$

(1 mark)

4 Solve.

$\log 2x = 5$

$x = \dots\dots\dots$

(2 marks)

- 5 A small tree is initially 2 metres tall.
Each year, the height of the tree increases by 20%.

The height, H metres, of the tree at the end of n years is given by the formula:

$$H = A \times b^n$$

- (i) Write down the value of A and b .

[2]

(ii) Find the exact time it takes for the tree to reach a height of 5 metres.

Give your answer in the form $n = \log_b a$.

[2]

(1 mark)

6 Find the value of x .

$$\log_3(4x + 1) = 4$$

(2 marks)

Hard Questions

1 $\log(5x + 2) = 2$

Find the value of x .

(2 marks)

2 Find the value of $\log_4\left(\frac{1}{16}\right)$

(2 marks)

3 The population of a village is 1000 people.

Each year, the population decreases by 10%.

The population at the end of n years is given by the formula

$$P = 1000 \times a^n$$

(i) Write down the value of a .

[1]

(ii) Find the exact time the population reaches 800.

Give your answer in an exact logarithm form.

[3]

(4 marks)

4 Make x the subject of this formula.

$$y = 10^x$$

(2 marks)

Very Hard Questions

1 $\log\left(\frac{1}{3x-2}\right) = -2$

Find the value of x .

(3 marks)

2 Find the value of $\log_3\left(\sqrt[4]{\frac{1}{9}}\right)$

(3 marks)

3 (a) A population of rabbits is increasing by the same amount each year.

The population, P , at the end of n years is given by the formula

$$P = 200 \times 2^n$$

(i) Write down the initial population, P_0 , of the rabbits.

[1]

(i) Write down the amount that the population increases by each year.

[1]

(2 marks)

(b) After how many years will the population first exceed 10 000?

(2 marks)

(c) Find the exact time the population becomes 200 000.

Give your answer in the form $n = \frac{\log b}{\log a}$.

(3 marks)